bloons

board

Board

heigth: int
width: int
Allballoons : List <Balloons>
grid : Cell[][]
#allTowers:List<Tower>
#evolutionTowers:List<Tower>
+PLAYER:Player

+Board(int,nt)
+*calculPath():void*
+getHeigth():int
+getWidth():int
+*getPath()* : List <Position>
+getCell(p : Position) : Cell
+displayBoard()
+displayPath()
+getGridCells() : Cell[][]
+getAllballoons() : List<Balloons>
+generateAllballons(int)
+processTower(List<Tower>, long)
+addTower(Tower)
+removeTower(Tower)
+EvoluteOneTower()
+removeEvolutionForOneTower()
+getAllTower
+getEvoludTower
+*createBalloonForPath(int, Level) : Balloon*
+*generateRandomPositionTower():List<Tower>*

Cell

- position:Positon

+ Cell(Position)
+getPosition():Positon

Position

-x:int
-y:int

+Position(int,int)
+getX():int
+getY():int
+equals(o:Object):boolean

Player

+PROFIT=10
+CREDIT=2500
+HEALTH=20
-health:int
-credit:int

+Player()
+getHealth():int
+getCredit():int
+addCredit()0
+addCredit(int)
+loseLife()
+removeCredit(int)
+canAfford(int)
+toString()

ClassicalBoard

path : List<Position>

+ClassicalBoard(int;int)
+calculPath():void
+getPath() : List <Position>
+isValid(Position) : boolean
+chooseValidNeighborPosition(Position) : Position
+createBalloonForPath(int, Level) : Balloon
+generateRamdomPositioTower():List<Tower>
+generateValidPosition():Position
+inPath(Position):boolean
+getValidTowerPosition():Position

RandomBoard

allPath : List<List<Position>>

+RamdomBoard(heigth:int,width:int)
+calculPath():void
+getPath() : List <Position>
+generatePath(int)
+getAllPaths() : List<List<Position>>
+getPathAt(int) : List<Position>
+displayBoard(List<Balloon>)
+createBalloonForPath(int, Level) : Balloon
+getValidTower():Position
+hasOverLap(List<Position>):Boolean
+generateRamdomPositioTower():List<Tower>

balloon

Balloon

- path : List<Position>
- position : Position
- level : Level
- health : int
- speed : double
- currentCellIndex : int
- prgresse : doubel

+Balloon (Position, Level)
+ getLevel() : Level
+ getPosition : Position
+ getSpeed() : double
+ getHealth() : int
+ getProgresse() : double
+ getCurrentCellIndex()
+ setHeath()
+ isDestroyed() : boolean
+ setTrajectory(List<Position>)
+ checkNextCell()
- updateBalloonPosition()
+ moveBalloon()
+ checkEndOfPath() : boolean
+ randomChoiceSpeed() : double
+ getPath() : List <Position>

utils

Direction <enum>

+ DOWN
+ RIGTH
+ TOP
- dx : int
- dy : int

+ Direction(int, int)
+ generateNeighbor(Position): Position

Level <enum>

+ NIV1
+ NIV2
+ NIV3
- resistance : int

- Level(int)
+ getResistance() : int
+ generateLevel() : Level

Projectile<enum>

+DART
+POINTEDDART
+VERYSHARPDART
+NEEDLE
+BOMB
+EXTRABOMB
-niveauDegat:int

-Projectile(int)
+getNiveau():int

Evolution<enum>

+RANGE
+ POWER
+CADANCE
+PROJECTILE

Effect<enum>

+ IMMEDIAT
+ DISTANCE

tower

projectileTower

SlowDownTower

+SlowDownTower(double,double,Effect,double,Position)
+attackBallon(Balloon,long)
+removeEvolution()
+putEvolution()

DartMonkeyTower

+DarkMokey(double,double,Effect,double, Positon,Projectile)
+attackBallon(Balloon,long):void
+removeEvolutionEffect(Evolution)
+applyEvolution(Evolution)

BombTower

+BombTower(double,double,Effect,double, Position,Projectile)
+attackBallon(Balloon,long):void
+removeEvolutionEffect(Evolution)
+applyEvolution(Evolution)

Tower

#cost: double
#rrange :double
#effect:Effect
#position: Position
#cadence: double
#timeLastAttack:long
#balloons: List<Balloon>
#balloonsInRange:List<Balloon>
#possibleEvolutions:Map<Evolution,Integer
appliedEvolution:Deque<Evolution>
+RANGE:double

+Tower(double,double,Effect,double, position)
+getEffect(): Effect
+getcadence():double
+ getRange(): double
+setRange(double)
+setCadence()
+getCost(): double
+getPosition():Position
+getAppliedEvolutions():Deque<Evolution>
+getPossibleEvolution():Map<Evolution,integer>
+inRange(Balloon):boolean
+calculDistance(Balloon):
+addBalloon(<ListBalloon>)
+getAllBalloonInRange:
+getAllBalloonInRange:
+updateAttackTime(long)
+getTimeLastAttack():long
+canAttack(long)
getBalloonInRange()
+toString():String
getBalloonClosetToExist():Balloon
+putEvolution()
+removeEvolution()
+*attackBallon(Balloon long)*
+*removeEvolutionEffect(Evolution)*
+*ApplyEvolution(Evolution)*

SniperMonkeyTower

+SnipeMonkeyTower(double,double, Effect,double,Position,Projectile)
+InRange(Balloon)
+removeEvolutionEffect(Evolution)
+applyEvolution(Evolution)

ProjectileTower

+projectile:Projectile

+ProjectileTower(double,double,Effect,double, Position,Projectile
+attackBallon(Ballon ,long):void
+getProjectile(Projectile):void
+setProjectile(Pojectile)
+removeEvolutionEffect(Evolution)
+applyEvolution(Evolution)

SuperMonkeyTower

+SuperMonkeyTower(double,double, Effect,double,Position,Projectile)
+removeEvolutionEffect(Evolution)
+applyEvolution(Evolution)

TackShooterTower

+TackShooterTower(double,double, Effect,double,Position,Projectile)
+removeEvolutionEffect(Evolution)
+applyEvolution(Evolution)

IceTower

-reelSpeed:double

+IceTower(double,double,Effect,double,Position)
+ freeze(Balloon)
+attackBallon(Balloon,long)
+removeEvolution()
+putEvolution()

action

BuyTowerAction

-tower:Tower
-position:Position

+BuyTowerAction(Tower,Position
+execute(Board):boolean
+getDescription:String
+getCost:int

Action<interface>

execute(Board):boolean
getDescription:String
getCost:int

UpgradeTowerAction

-tower:Tower

+UpgradeTowerAction(Tower)
+execute(Board):boolean
+getDescription:String
+getCost:int

DowngradeTowerAction

-tower:Tower

+DowngradeTowerAction(Tower)
+execute(Board):boolean
+getDescription:String
+getCost:int